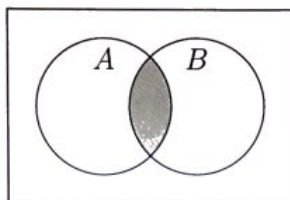
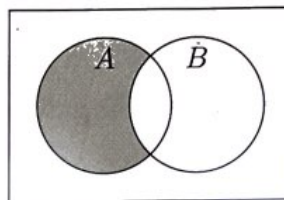


4.16 Classwork: Probability Review

1. (a) For each Venn diagram below, write an expression for the shaded region.

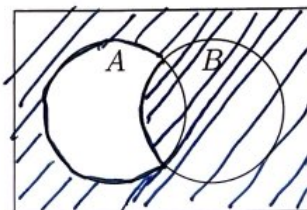


$$A \cap B$$



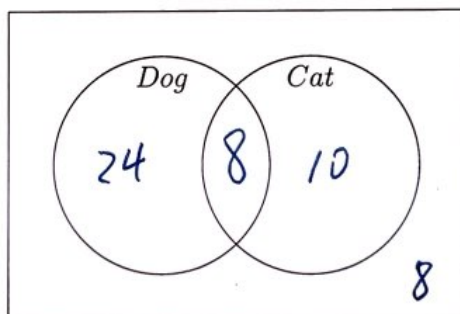
$$A \cap B'$$

- (b) Shade the region representing $A' \cup B$ in the diagram below.



2. In a survey of 50 students, 32 own a dog, 18 own a cat, and 8 own both.

- (a) Complete the Venn diagram below, writing the number of students in each region.



- (b) A student is selected at random. Find the probability they own a dog but not a cat.

$$P(D \cap C') = \frac{24}{50}$$

- (c) Find the probability they own neither a dog nor a cat.

$$P(D' \cap C') = \frac{8}{50}$$

3. Events A and B are such that $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$.

(a) Find $P(A \cup B)$. (Show substitution into a formula.)

$$\begin{aligned} P(A \cup B) &= 0.6 + 0.5 - 0.2 \\ &= 0.9 \end{aligned}$$

(b) Find $P(A' \cap B')$.

$$P(A' \cap B') = \cancel{0.9} 1.0 - 0.9 = 0.1$$

(c) Find $P(A | B)$. (Show substitution into a formula.)

$$P(A|B) = \frac{0.2}{0.5} = \frac{2}{5}$$

(d) Determine whether A and B are independent. Justify your answer.

$$\begin{aligned} \text{No. } P(A) \cdot P(B) &= 0.6 \times 0.5 \\ &= 0.3 \neq P(A \cap B) \end{aligned}$$

Not independent

4. Events A and B are independent, with $P(A) = 0.4$ and $P(B) = 0.7$.

(a) Find $P(A \cap B)$.

$$P(A) \cdot P(B) = 0.4 \cdot 0.7 = 0.28$$

(b) Complete the probability table below, including all totals.

	A	A'	Total
B	0.28	0.42	0.7
B'	0.12	0.18	0.3
Total	0.4	0.6	1.0

(c) Find $P(A \cup B)$.

$$= 0.4 + 0.7 - 0.28$$

$$= 0.82$$

(d) Find $P(B | A')$.

$$= \frac{P(B \cap A')}{P(A')} = \frac{0.42}{0.6} = 0.7$$

5. Events A and B are mutually exclusive, with $P(A) = 0.35$ and $P(B) = 0.25$.

(a) Write down $P(A \cap B)$.

0

(b) Find $P(A \cup B)$.

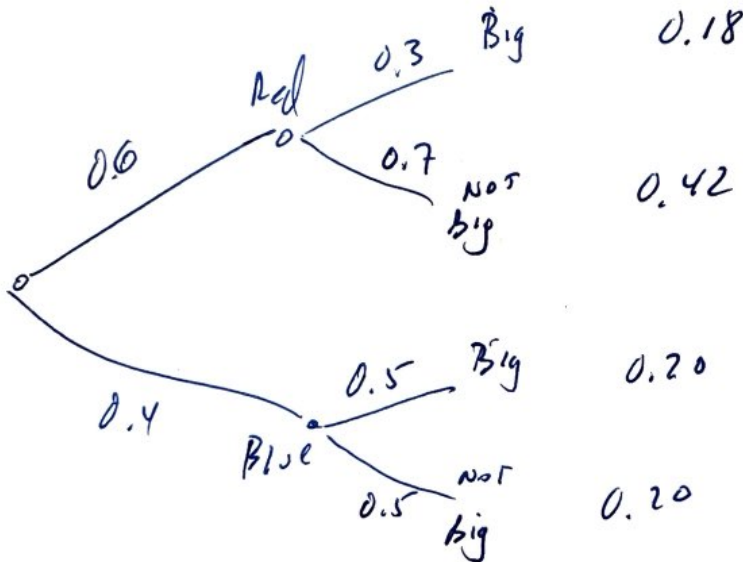
$$0.35 + 0.25 = 0.6$$

(c) Find $P(A' \cap B')$.

$$= 1.0 - 0.6$$

6. In a bag of marbles, 60% are red and 40% are blue. When a red marble is selected, the probability that it is large is 0.3. When a blue marble is selected, the probability that it is large is 0.5. A marble is selected at random.

(a) In the space below, draw a fully labelled tree diagram for this situation, including all branch probabilities.



(b) Find the probability that the marble is red and large.

$$0.6 \cdot 0.3 = 0.18$$

(c) Find the probability that the marble is large.

$$0.18 + 0.20 = 0.38$$

(d) Given that the marble is large, find the probability that it is red.

$$P(R/L) = \frac{0.18}{0.38} \quad \left(= 0.47368... \right)$$

$$\quad \quad \quad \left(\approx 0.474 \right)$$